Electricity from coal? 18a

These days we hear quite a lot about building energy-efficient houses and insulating houses in order to reduce the amount of electricity we use. We are also being educated about how we can use less electricity at home. There is also continuing research into alternative sources of electricity, although most of our electricity is produced from coal. Should we continue producing electricity from coal? In this activity you are going to discuss the advantages and disadvantages of coal-electricity and then draw your own conclusion.

The following information about electricity production from coal has been collected and placed into three boxes.

A. Explanation of coal-electricity:

- Black coal is burnt.
- Black coal is mined.
- Heat boils water into steam.
- Steam turns a turbine.
- Turbine changes movement into electricity.

B. Arguments for coal-electricity:

- Australia exports about half of its coal to Japan and Europe.
- Coal can be mined in all weather conditions.
- Coal deposits are near the surface of the earth.
- Easy to extend network to new areas.
- Electricity from coal generators is reliable.
- Electricity is not dependent on weather or cycles.
- Enough coal for 400 years of electricity production.
- Established network of coal generators and electric cables.
- Large coal deposits in Australia.
- Network is working well at present.
- No tunnels and lifts needed as in underground mines.
- Open-cut mining is cheap.
- Open-cut mining is easy.

C. Arguments against coal-electricity:

- Acid rain damages plants and weathers buildings.
- Alternative ways to produce electricity need to be researched, which is expensive.
- It is expensive to test and improve alternative electricity production methods.
- Changes in temperature and chemicals disturb the natural balance of the environment.
- Coal burning is responsible for nearly 50% of Australia's carbon dioxide emission.
- Coal is washed with water from lakes and rivers.
- Dirty water goes back but is now polluted with chemicals.
- Global warming will change weather patterns and raise the ocean levels.
- Habitats are reduced and partly destroyed.
- Increased carbon dioxide in the atmosphere leads to global warming.
- Large areas of land are necessary for mining of coal.
- Many natural bush areas and fields are used up for electric cables and electric towers.
- Much land is used for coal power stations.
- Nitrogen oxides and sulfur dioxide from burning coal dissolve in raindrops.
- Non-renewable resource will run out eventually.
- Power plant equipment is cooled with river and lake water.
- Raindrops contain nitric acid and sulfuric acid.
- 1 MWh (= megawatt hour) of electricity produces nearly 1 ton of carbon dioxide.

1 Arrange the points into order by writing numbers in front of them: box A is about how electricity is produced from coal; box B provides four main arguments in favour of coal-electricity; box C contains five main arguments against coal-electricity.

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Electricity from coal? 18b

2 You are now going to write your discussion. First introduce the issue of coal-electricity and explain it. Then make it clear that there are arguments in favour of coal-electricity and arguments against it. Write the introductory paragraph, referring to your first group of ordered pieces of information. Use the scaffold below and the hints at the end to help you.

Introduction:	
Statement of issue	
Explanation of issue	
Preview of arguments	

Now write the section supporting coal-electricity. There are four arguments in its favour. For each argument there should be a main point (ie *what is this argument about?*) and an elaboration (ie *what explains or supports this argument?*). Use your ordered statements from box B above and the scaffold below to write four arguments in favour of coal-electricity.

Arguments for:	
Point 1: Size of coal deposits Elaboration 1	
Point 2: Open-cut mining Elaboration 2	
Point 3: Reliable electricity Elaboration 3	
Point 4: Network of coal-electricity Elaboration 4	

Now write the section which is against coal-electricity. There are five arguments against it. Again use the structure of point and elaboration to construct each argument. Use your ordered statements from box C above and the scaffold below to write five arguments against coal-electricity.

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Arguments against:	
Point 1: Non-renewable resource Elaboration 1	
Point 2: Greenhouse gas emission Elaboration 2	
Point 3: Acid rain gas emission Elaboration 3	
Point 4: Destruction of land areas Elaboration 4	
Point 5: Thermal and chemical pollution Elaboration 5	

Finally, you must write a conclusion. Decide whether you are in favour of coal-electricity or against it, and write down your opinion. Note that there is no right or wrong answer, but it is important that you are able to justify your point of view. Justify your opinion by summarising the most important arguments. To finish, you may like to recommend what you think ought to happen in the future. In your conclusion you need to sound convincing: use emotive words (eg *terribly important, widespread devastation, must, should never*) and exaggerate a little (use words like *always, rarely, obviously, everybody*). Write your conclusion in the scaffold below.

Conclusion:	
Opinion	
Summary of key arguments	
Recommendation	

Handy hints:

- Statement of issue = what is this discussion about?
- Explanation of issue = how is electricity produced from coal?
- Preview of arguments = what are the arguments about?
- Point = what topic is this paragraph about?
- Elaboration = what information is there to support or explain the topic?
- Opinion = which side do you think is right?
- Summary of key arguments = what are the most important arguments supporting your view?
- Recommendation = what should happen now?

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